

PROJECT SUMMARY · PREPARED FOR THE CEO

WeLearn Growth Hub

Platform walkthrough, feature reference, and the plan to enterprise release

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Date: 11 September 2026 | **Environment:** dnd-master-73449.web.app

Stack: Vanilla JS + Vite, Firebase Authentication & Firestore, Google Gemini

Status: Working prototype, tested end to end — awaiting a first classroom pilot

1. Executive Summary

WeLearn Growth Hub is a gamified STEM learning platform for students aged 12–18, paired with a monitoring portal for the coaches who teach them. Students set their own project goals with AI guidance, work through five interactive climate and engineering missions, and track progress daily. Coaches see the whole class in one view, rate work, leave feedback, set assignments, and export progress reports.

Both halves are live and functioning. A defect review during this period closed a privilege-escalation flaw, four features that reported success to the user while silently failing to save, and two charts that displayed invented figures where data was missing. Every fix was confirmed against the running system rather than inferred from a successful build.

5

INTERACTIVE STEM MISSIONS

0

CLASSROOM PILOTS RUN TO DATE

8

DEFECTS CLOSED THIS PERIOD

10

WEEKS TO ENTERPRISE RELEASE

One point of context matters for reading everything that follows: **no real class has used the platform yet.** Every account created to date is a test account, so the features described here are demonstrated rather than proven in a classroom. The remaining gap is therefore not capability but *evidence and assurance*. The plan in Section 7 puts a genuine pilot first, then widens reach — mobile devices, the Thai language — and finally puts testing and compliance controls around the platform before it holds real student records.

2. How the Platform Works

One web address serves both audiences. A single sign-in routes each person to the right experience based on the role held against their account.

- 1 Sign in.** The visitor chooses Student Portal or Coach Portal. Credentials are handled by Firebase Authentication; the platform never stores passwords itself.

- 2 **Role routing.** The application reads the account's role and presents either the student dashboard or the coach dashboard. Roles cannot be self-assigned.
- 3 **Goal setting.** A student new to the platform is guided through a four-step wizard establishing what they will learn, on what topic, over what period, and which specific goals they are committing to.
- 4 **Learning.** The student works through the five missions, with an AI companion available throughout for hints, explanations and accessibility support.
- 5 **Monitoring.** Coaches see every student in their classes with live progress, goals, time on task and project telemetry.
- 6 **Intervention.** Coaches rate work, write feedback, and send notifications that reach the student on their own dashboard.

3. The Student Experience

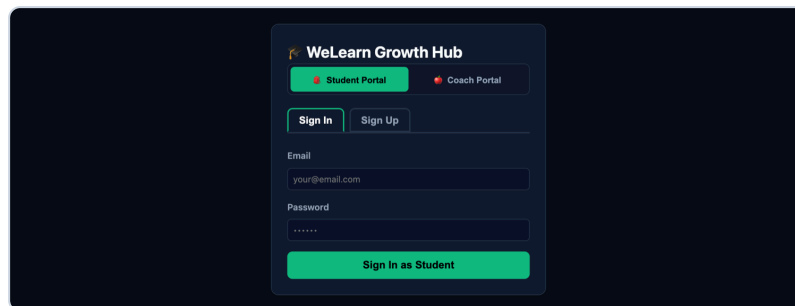


Figure 1 — One entry point. The student or coach choice determines everything that follows.

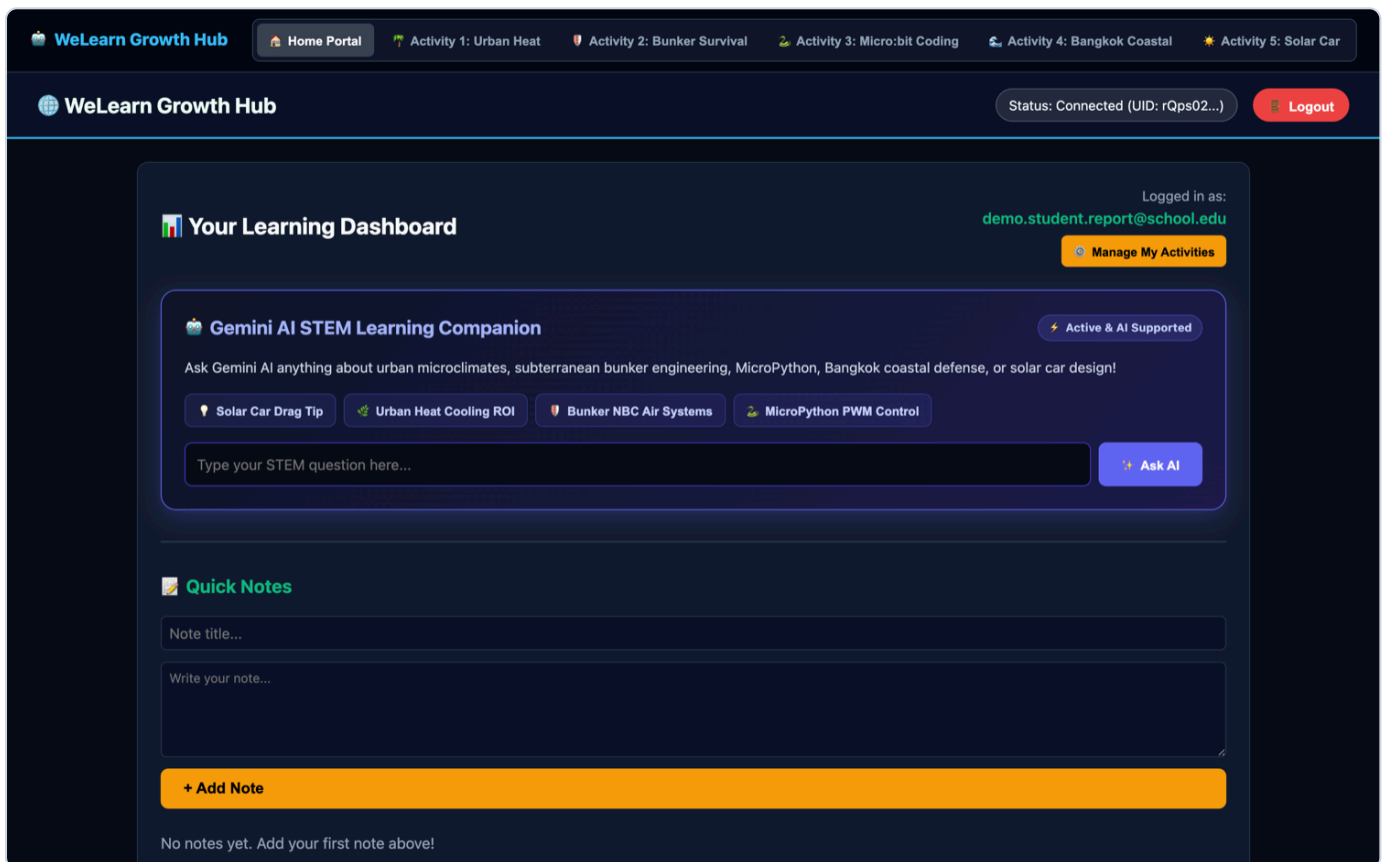


Figure 2 — The student dashboard. Navigation across the top gives direct access to goal setting and all five missions.

Goal-setting wizard

A four-step guided flow: choose a learning path (climate or coding), pick a project topic from a curated list, set a project length, then write the specific goals to be achieved. An AI helper comments at each step, so a student who is unsure is not left staring at a blank field.

Reachable any time from the "Goal Setting" tab, not only at first sign-in.

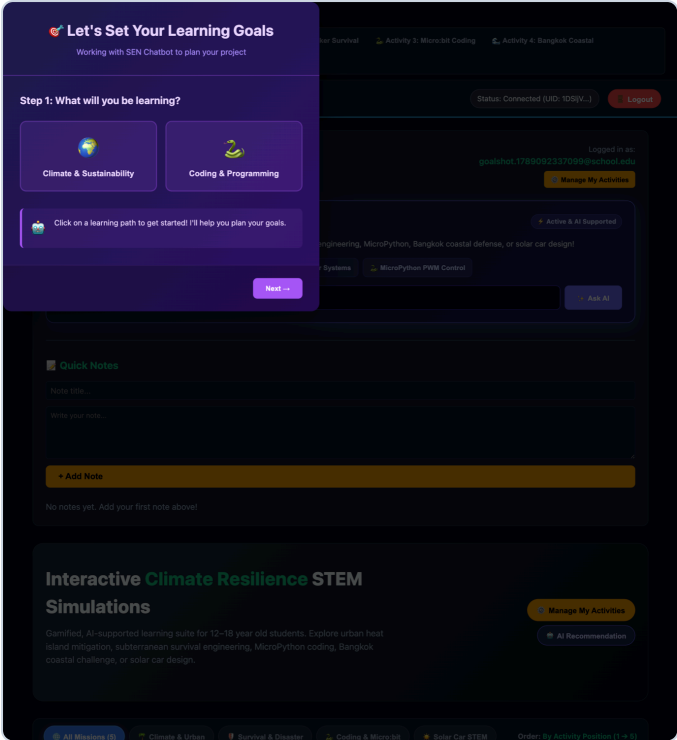


Figure 3 — Step 1: learning path.

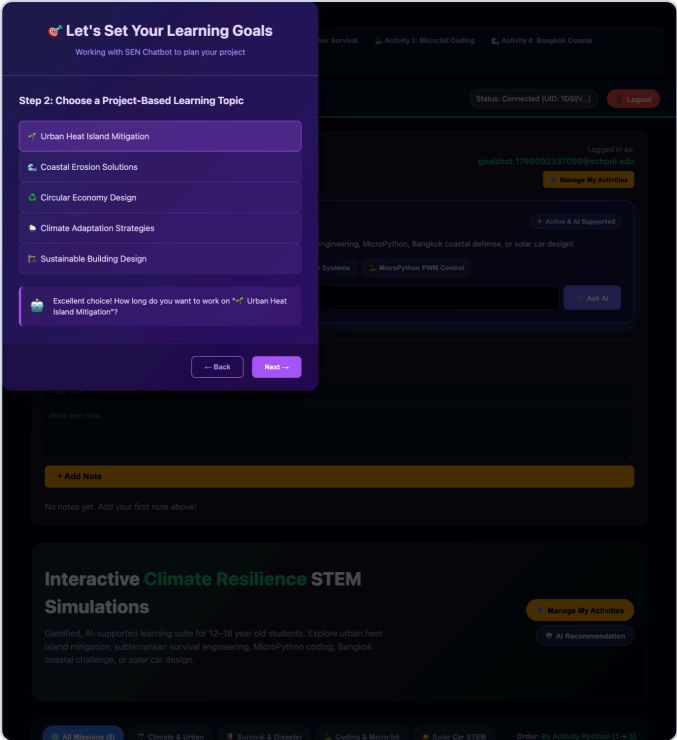


Figure 4 — Step 2: project topic.

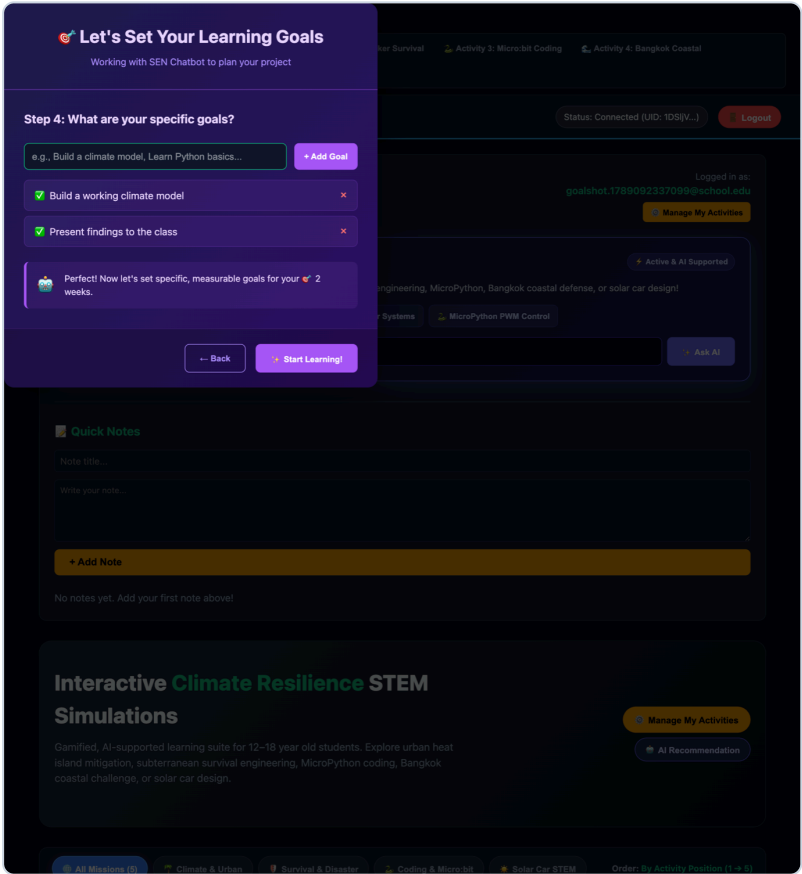


Figure 5 — Step 4: the student writes their own goals, which become their checklist.

Learning Project Goals

Once set, goals become the student's home base: a progress percentage, a countdown of days remaining to the project deadline, and a tick-box checklist of each goal. Editing is one click away via "Set / Edit Goals".

Goals are stored centrally, so they follow the student to any device — and the same records are what the coach sees.

Daily Goal & Focus Reminder

A standing prompt showing days remaining and the project deadline, with two actions: "Mark Today's Learning Done" to record the day, or "Snooze 24h" to defer it. The intent is a light daily habit rather than a notification that gets ignored.

Gemini AI STEM Learning Companion

An always-available assistant for questions about any mission — urban microclimates, bunker engineering, MicroPython, coastal defence, solar car design. Four one-tap prompts cover the most common questions; free text handles the rest.

Accessibility support

Detects when a learner appears stuck and offers help unprompted, rather than requiring them to recognise they need it and ask. Built for students with additional learning needs, useful to everyone.

Quick Notes and personalisation

A private notebook attached to the account, plus "Manage My Activities" to choose which missions appear and an AI recommendation for what to attempt next.

4. The Five Missions

WeLearn Growth Hub

Home PortalActivity 1: Urban HeatActivity 2: Bunker SurvivalActivity 3: Micro:bit CodingActivity 4: Bangkok CoastalActivity 5: Solar Car

WeLearn Growth Hub

Status: Connected (UID: rQps02...)Logout

Your Learning Dashboard

Logged in as:
demo.student.report@school.edu

Manage My Activities

Gemini AI STEM Learning Companion

Active & AI Supported

Ask Gemini AI anything about urban microclimates, subterranean bunker engineering, MicroPython, Bangkok coastal defense, or solar car design!

Solar Car Drag TipUrban Heat Cooling ROIBunker NBC Air SystemsMicroPython PWM Control

Type your STEM question here...

Ask AI

Quick Notes

Note title...

Write your note...

+ Add Note

No notes yet. Add your first note above!

Interactive Climate Resilience STEM Simulations

Manage My Activities

AI Recommendation

Gamified, AI-supported learning suite for 12–18 year old students. Explore urban heat island mitigation, subterranean survival engineering, MicroPython coding, Bangkok coastal challenge, or solar car design.

All Missions (5)Climate & UrbanSurvival & DisasterCoding & Micro:bitSolar Car STEM

Order: By Activity Position (1 → 5)

POSITION #1

BeginnerAI Supported

ACTIVITY 1 - URBAN SIMULATION

Urban Heat Battle

Model tropical urban heat island mitigation in Townsville (19°S). Test green roofs, mature street trees, water features, and vertical green walls to cool city temperatures.

Green Roofs & Albedo CoatingsStreet Tree Canopy (High ROI)Water Features & Permeable PavingReal-time UTCI Thermal Index

Launch Activity 1: Urban Heat

Ask AI Tutor About Activity 1

POSITION #2

IntermediateAI Supported

ACTIVITY 2 - SURVIVAL ENGINEERING

Bunker Survival

Engineer a subterranean bunker to withstand 50°C heatwaves, Category 5+ hypercanes, toxic wildfires (AQI 999+), and mega-floods based on Bomb Shelter Supply specs.

Subterranean Steel & Concrete HullsSwiss NBC/CBRN Positive Air ScrubbersAtmospheric Water & Deep WellsSolar, Diesel & Geothermal HVAC

Launch Activity 2: Bunker Survival

Ask AI Tutor About Activity 2

POSITION #3

BeginnerAI Supported

ACTIVITY 3 - MICROPYTHON SIMULATOR

Micro:bit Coding

Learn MicroPython by controlling virtual 5x5 LED displays, servo motors, and radio direction sensors. Perfect for neurodivergent STEM learners

POSITION #4

IntermediateAI Supported

ACTIVITY 4 - COASTAL DISASTER CHALLENGE

Bangkok Coastal Challenge

Decode orbital binary alert signals, convince fisherman Somchai to evacuate, select the optimal coastal bunker, and calculate storm surge

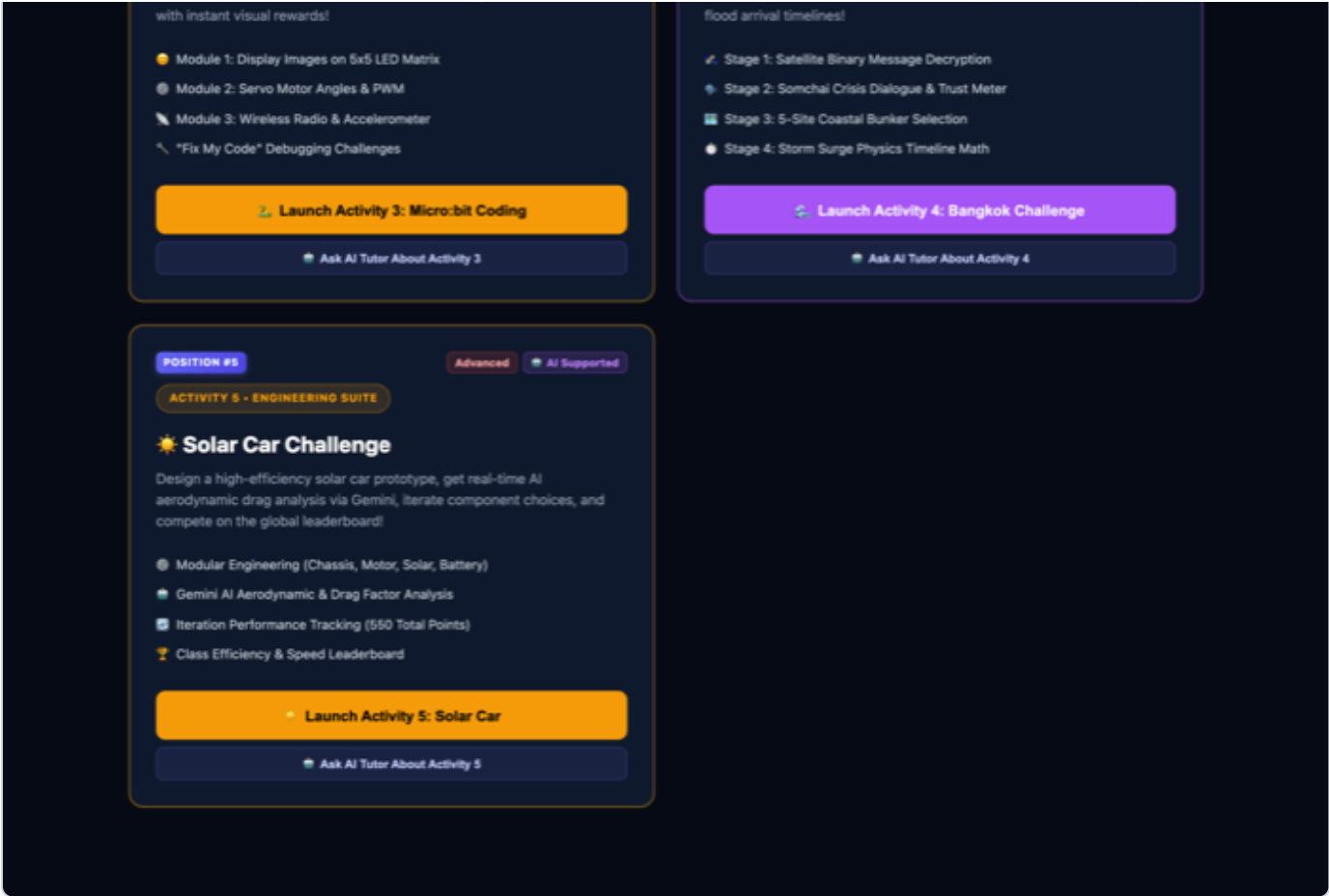


Figure 6 — The mission catalogue. Each card carries its curriculum position, difficulty, an AI-supported badge, and what the student will actually do.

Mission	Level	What the student does
Urban Heat Battle	Beginner	Models heat-island mitigation for a tropical city — green roofs, street tree canopy, water features, permeable paving — against a live thermal comfort index and a fixed budget.
Bunker Survival	Intermediate	Engineers a subterranean shelter against 50°C heatwaves, Category 5+ storms, toxic wildfire smoke and flooding: structure, air filtration, water, power and supplies.
Micro:bit Coding	Beginner	Writes real MicroPython against a simulated micro:bit — LED matrix, servo motors and PWM, wireless radio and accelerometer — plus "fix my code" debugging challenges.
Bangkok Coastal Challenge	Intermediate	A four-stage crisis: decode a satellite binary alert, persuade a reluctant fisherman to evacuate, choose the best shelter site, then calculate storm-surge arrival times.
Solar Car Challenge	Advanced	Designs a solar vehicle from modular components, receives AI aerodynamic and drag analysis, iterates across versions, and competes on a class leaderboard.

The sequence is deliberate — beginner modelling, then engineering, then coding, then applied crisis response, then open-ended design. Coaches can reorder it per class from the "Activities Position & Curriculum" view.

5. The Coach Experience

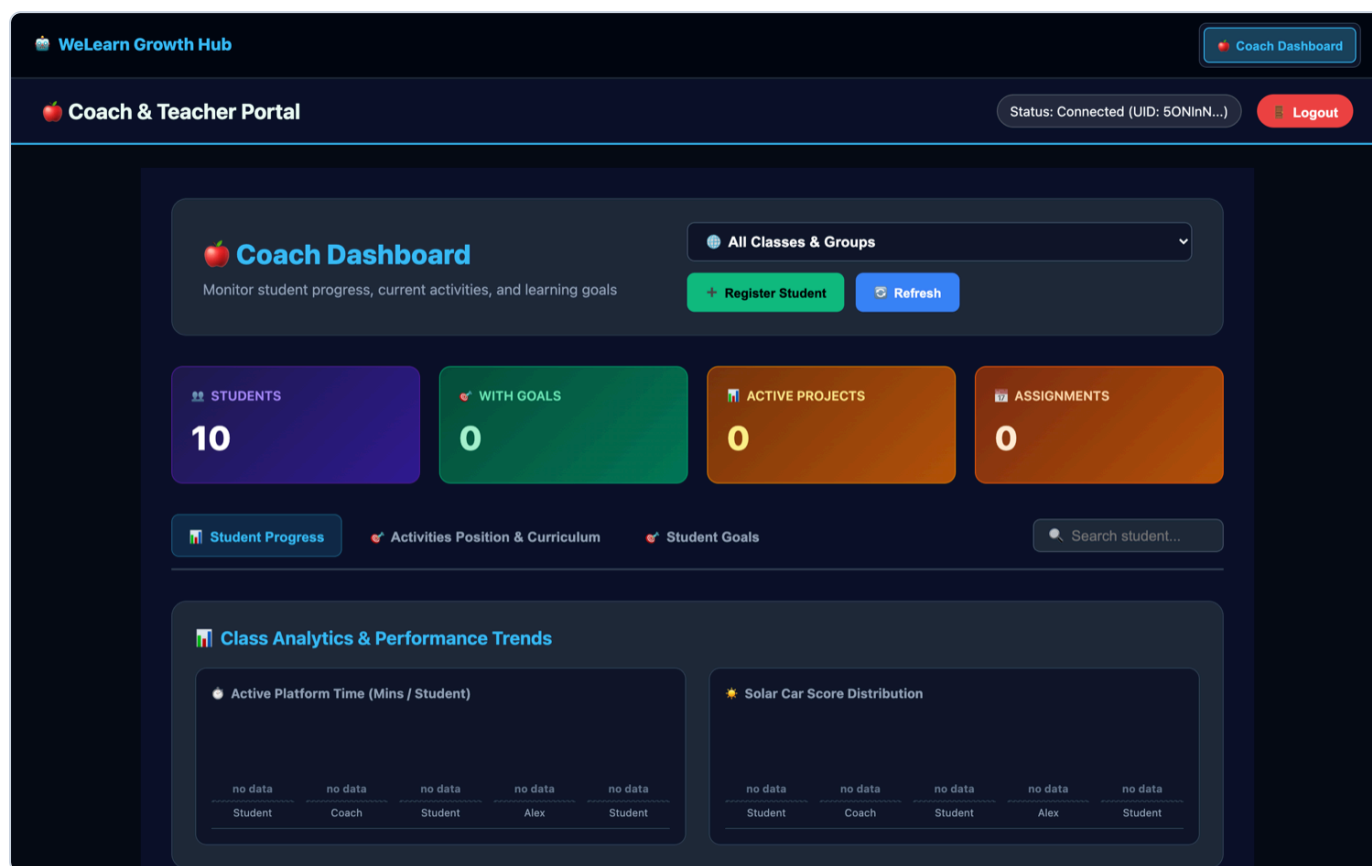


Figure 7 — The coach dashboard: class-wide counts, three views, and every student below.

Class overview

Four headline counts — students, students with goals set, active projects, and open assignments — filtered by class or group. "Register Student" enrolls a learner directly; "Refresh" re-reads live data.

Three views

Student Progress — every student with their work and metrics. **Activities Position & Curriculum** — the order missions are presented in. **Student Goals** — goals across the class with status, for comment. A search box filters by student name.

Class analytics

Active platform time per student and solar car score distribution. Where a student has no recorded data the chart says so explicitly rather than substituting a placeholder figure — a coach must never mistake an invented number for a real one.

Student cards and work rating

Each student shows their group, points, project telemetry (iteration, weight, motor) and total time spent, with a five-star control to rate the quality of their work. Ratings save to the student's record.

Student detail and coach feedback

"View Details & Provide Feedback" opens full solar car telemetry and specifications, the student's goals, and a free-text box. Feedback appears directly on that student's dashboard.

Assignments and progress export

Coaches set assignments with deadlines against a class, and export a fourteen-column progress report — activities started and completed, goals by status, time on task, active days, last-active date and engagement level — for reporting or parent communication.

6. Platform Status

Capability	Status	Note
Sign-in and role routing	Working	Roles are server-controlled and cannot be self-assigned.
Goal-setting wizard	Working	Four steps; reachable at any time.
Goals, checklist and daily reminder	Working	Stored centrally and visible to the coach.
Five STEM missions	Working	All launch and run.
AI companion and accessibility support	Working	Available throughout.
Coach dashboard and class metrics	Working	Load time reduced from 6.1s to 3.3s.
Ratings, feedback and notifications	Working	Verified saving end to end.
Assignments with deadlines	Working	Coach-authored, student-readable, forgery-resistant.
Progress export	Working	Fourteen columns.
Recorded learning activity	None yet	All accounts to date are test accounts. No class has used the platform, so no genuine learning data exists and the analytics have nothing to describe.
Class analytics charts	Partial	Two of the four planned charts.
Thai language	Not built	A translation module exists but is not yet connected to any screen.
Mobile and tablet layouts	Not built	Breakpoints exist but are unverified on real devices.
Automated test coverage	Not built	The absence of tests is why the defects above went undetected.

7. Operating Cost

Estimated running cost for a single cohort of up to 30 students, based on the platform's observed behaviour. These are infrastructure and service costs only; they exclude people.

Service	Monthly	Basis
Authentication	\$0	Free to 50,000 monthly active users.

Database	\$0–3	~10,000 reads and 400 writes per day against a 50,000 read / 20,000 write daily allowance.
Hosting and SSL	\$0–2	~31 MB per day against a 360 MB daily allowance.
AI (Gemini)	\$2–15	Roughly 4.7M input and 2.7M output tokens per month across all students.
Server-side AI proxy	\$0–5	Within the free invocation allowance at this volume.
Custom domain	~\$1.50	Annual registration, amortised.
Total	\$5–25	Central estimate \$10–15 per month, or \$150–200 per year.

Two cost decisions worth taking deliberately

Use the paid AI tier, not the free one. Free-tier terms have historically permitted prompts to be used for model improvement. With minors' schoolwork passing through the service that is the wrong trade at any price — and at these volumes the paid tier costs single digits per month, so there is no meaningful saving to defend.

Cost scales with class size, not linearly. The coach dashboard reads every student's records on each refresh. At 30 students this consumes about 1% of the free allowance. At 500 students across a full staff room it becomes both slow and billable. That is addressed in the Week 3 performance work and is a future concern, not a present one.

These figures are estimates derived from how the application behaves, not measurements of real classroom use. A month of pilot data would convert the range into a defensible number.

8. Ten-Week Plan to Enterprise Release

Sequenced so that real usage data exists before analytics are built on top of it, reach is widened before the platform goes in front of a whole school, and assurance is in place before it holds student records at scale.

Week	Workstream	Outcome
Week 1	Credential hygiene	Rotate the exposed AI key at the provider, purge it from repository history, and move AI calls behind a server-side proxy so no key ever reaches the browser.
Week 2	Activity telemetry	Every mission records progress, scores and time on task as the student works. Without this the coach portal has little to report, which is its single largest limitation today.
Week 3	Pilot with a real class	One class runs end to end for a week. Produces genuine data and surfaces the workflow problems that only appear in a classroom.
Week 4	Coach portal depth	Individual student drill-down with intervention flagging, assignment submission and grading, and class-versus-class comparison.
Week 5	Analytics	Completion trend over time and class-versus-cohort benchmarking added to the two existing charts. PDF and Excel export.
Week 6	Mobile and tablet	Verified layouts at 320/768/1024/1440px on real devices, with touch targets and mobile navigation — the devices students actually use.
Week 7	Accessibility	WCAG AA conformance: contrast, keyboard navigation, screen-reader labelling. Independently audited.
Week 8	Thai language	Full interface in Thai with locale-aware dates and numbers, reviewed by a native speaker.
Week 9	Testing and compliance	Automated tests above 80% coverage, covering every save path. Audit logging, data retention and access records for FERPA alignment.
Week 10	Launch	Coach acceptance testing, load testing at full school scale, staging sign-off, training materials, production release.

Definition of done

- No credential in source control or reachable from the browser
- Every student action recorded; every coach action verified saving under test
- Dashboard interactive within 3 seconds at full school scale
- WCAG AA confirmed by independent audit; Lighthouse above 85 across all categories
- Full interface available in Thai

- Ten teachers complete acceptance testing without a blocking defect

Assumptions

- Two full-stack engineers, with part-time QA and design from Week 4
- A class and its teacher are available for the Week 3 pilot
- A Thai-speaking reviewer is available in Week 8
- Teacher availability is secured in advance for Week 10 acceptance testing

Recommendation

Approve Weeks 1 to 3 as a single block. They convert the platform from something that works into something with evidence behind it: credentials secured, real learning activity recorded, and one class having actually used it. That evidence should then inform whether the remaining seven weeks are worth the investment — a far better basis for that decision than this document alone.

WeLearn Growth Hub — Project Summary, 11 September 2026. Feature status confirmed by direct observation of the running application. Screenshots are unmodified captures of the live platform.